

SULTRIM SUSPENSION

Ingredient(s):
Each ml contains :
Sulfadiazine 400mg (potency)
Trimethoprim 80mg (potency)

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):
Pharmacodynamics
Sulfadiazine -
The sulfonamides are bacteriostatic antimicrobials that interfere with the biosynthesis of folic acid in bacterial cells; they compete with para-aminobenzoic acid (PABA) for incorporation into dihydrofolic acid. By replacing the PABA molecule in dihydrofolic acid, they prevent formation of folic acid required for nucleic acid synthesis and multiplication of the bacterial cell. Sulfonamides are effective only in cells that must produce their own folic acid; mammalian cells do not synthesize folic acid, but get it from outside sources.
Trimethoprim –
Trimethoprim belongs to diaminopyrimidines, and is a bacteriostatic antimicrobial that blocks a step in folate production just subsequent to that affected by the sulfonamides. Bacterial production of tetrahydrofolic acid from dihydrofolate is interrupted by the diaminopyrimidine as it reversibly binds and inhibits dihydrofolate reductase. Because the conversion of dihydrofolic acid to tetrahydrofolic acid is blocked, folate cannot be produced. Pyrimidine causes the same inhibition of dihydrofolate reductase in protozoa. Like bacteria and protozoa, animal cells also reduce folic acid to tetrahydrofolic acid; however, bacterial and protozoal dihydrofolate reductase is significantly more tightly bound by trimethoprim than is human dihydrofolate reductase.
Potentiated sulfonamides (Sulfadimethoxine + Trimethoprim) –
The potentiated sulfonamides are the combination of diaminopyrimidines and sulfonamides. Because the diaminopyrimidines exert their effect on folate biosynthesis at a step immediately subsequent the one at which the sulfonamides act, the combination of a sulfonamide and diaminopyrimidine produces a synergistic effect that deprives the cell of essential nucleic acids and proteins. The potentiated sulfonamide combination produces an antimicrobial effect that is bacteriostatic and sometimes bactericidal against certain bacteria under optimum conditions.

Pharmacokinetics
Sulfadiazine -
The absorption of oral sulfadiazine in calves is slow but complete and unaffected by rumen status. The sulfonamide component generally is found at higher concentrations in plasma for a much longer time and tissue distribution is slower. Initial concentrations of sulfonamides in tissues are generally lower than those in plasma. The binding of sulfonamides to protein is concentration-dependent. In cattle, protein binding is approximately 50%. Sulfonamides are metabolized primarily in the liver, but metabolism also occurs in other tissues. Biotransformation occurs by acetylation, glucuronide conjugation, and aromatic hydroxylation in many species. The types of metabolites formed and the amount of each varies depending on the specific sulfonamide administered; the species, age, diet, and environment of the animal; the presence of disease; and, with the exception of pigs and ruminants, the gender of the animal. N4-acetyl metabolites have no antimicrobial activity and hydroxymetabolite have 2.5–39.5% of the activity of the parent compound. Metabolites may compete with the parent drug for involvement in folic acid synthesis. They have little detrimental effect on the bacterial cell, so their presence could decrease the activity of the remaining parent drug. Renal excretion is the primary route of elimination for most nonenteric sulfonamides and it occurs by glomerular filtration of parent drug, tubular excretion of unchanged drug and metabolites, and passive reabsorption of nonionized drug. Alkalization of the urine increases the fraction of the dose that is eliminated in the urine. In general, the metabolites of the parent drug are more quickly eliminated by the kidney than is the original sulfonamide, but the proportions of metabolites formed can vary depending on many factors. In calves, the percentage of unchanged drug excreted increases from 1 day of age to 42 days of age, changing from 22–50%.
Trimethoprim –
The bioavailability of oral trimethoprim is greatly reduced in ruminating calves because it is apparently trapped in the rumenoreticulum and undergoes some degradation. In horses, the absorption of trimethoprim is delayed when free access to feed and initial serum concentrations will be lower in a fed horse than in a fasted horse. In general, the diaminopyrimidine concentration in plasma peaks early and is quickly found in high concentrations in tissues than in the serum. In mares, trimethoprim was measured in CSF at 50% of serum concentrations after 4-day dosing. Protein binding is independent of plasma concentration with approximately 57% in cows, 48% in goats, 50% in horses and 33–54% in pigs. In many species, including cows, goats, and pigs, trimethoprim is extensively metabolized to oxide and hydroxylated metabolites. Serum elimination half-lives are 2.5 hours in dogs, 1.91–3 hours in horses, 1.5 hours in cattle, and 2.4 hours in pigs. Renal excretion of trimethoprim occurs by glomerular filtration, active tubular secretion, and reabsorption. In dogs, two-third of the total dose is eliminated in the urine as parent drug in the first 24 hours. In horses, it is believed that a large percentage of trimethoprim is metabolized before elimination in urine (46%) and feces (52%). The clearance of trimethoprim is affected by urine pH, plasma concentration, and extent of diuresis.

Potentiated sulfonamides (Sulfadimethoxine + Trimethoprim) –
Trimethoprim/sulfa is well absorbed after oral administration, with peak levels occurring about 1–4 hours after dosing. Trimethoprim/sulfa is well distributed in the body. When meninges are inflamed, the drugs enter the CSF in levels of about 50% of those found in the serum. Both drugs cross the placenta and are distributed into milk. Trimethoprim/sulfa is both renally excreted unchanged via glomerular filtration and tubular secretion and metabolized by the liver.

Indication(s):
Bacterial infections due to susceptible organisms in chicken, and turkey.

Target Species
Poultry.

Dosage and Administration:
Slowly add measured volume to water in medicated tank, stirring thoroughly.
Escherichia coli, *Pasteurella*, *Haemophilus* and staphylococcal infections in chicken: 1ml/4.5L water.
Salmonella infections in chickens: 3ml/4.5L water.
Bacterial infections in turkey: 15ml/500kg flock liveweight in water consumed daily.
Recommended dilution rates apply when water consumption is normal; if subnormal, increase concentration to administer 15 to 30mg/kg total liveweight daily (active ingredients). Continue treatment for 3 to 5 days according to severity of disease.

Contraindications:
Except under special circumstances, this medication should not be used when the following medical problems exist:
1) Blood dyscrasias: Slight to moderate reduction in hematopoietic activity has been reported with long-term high dosing of potentiated sulfonamides.
2) Hypersensitivity to diaminopyrimidines or sulfonamides: Animals that have had a previous reaction may be much more likely to react on subsequent administration.

Precaution(s) / Warning(s):
1) Trimethoprim/sulfa should not be used in dogs or horses showing marked liver parenchymal damage, blood dyscrasias, or in those with a history of sulfonamide sensitivity.
2) An idiosyncratic sulfonamide toxicosis can occur in any breed of dog, but the reaction has been reported more frequently in the Doberman Pinscher than in other breeds. This specific type of drug reaction includes blood dyscrasias, nonseptic polyarthritis, and skin rash.
3) Risk-benefit should be considered when the following medical problems exist:
a. Hepatic function impairment: Delayed biotransformation may increase the risk of adverse effects.
b. Renal function impairment: Delayed elimination could cause accumulation of sulfonamide and metabolites, increasing the risk of adverse effects.
c. Urolithiasis: Sulfonamides can crystallize in the renal system under certain conditions.

Interaction with Other Medicaments
1) Trimethoprim/sulfa may prolong the clotting times in patients receiving coumarin (warfarin) anticoagulants.
2) Sulfonamides may displace other highly bound drugs such as methotrexate, phenylbutazone, thiazide diuretics, salicylates, probenecid and phenytoin.
3) Antacids may decrease the bioavailability of sulfonamides if administered concurrently.
4) Trimethoprim may decrease the therapeutic effect of cyclosporine (systemic) and increase the risk of nephrotoxicity developing.
5) A trimethoprim and sulfonamide combination administered to a detomidine-anesthetized horse can lead to arrhythmias, hypotension, and death; it is suspected that the antimicrobial potentiates the cardiac changes reported with detomidine.

Pregnancy and Lactation
Sulfonamides and diaminopyrimidines cross the placenta in pregnant animals and some teratogenic effects have been seen with very high doses given to pregnant mice and rats.

Side Effect(s) / Adverse Reaction(s):
1) Adverse effects noted in dogs: keratoconjunctivitis sicca (which may be irreversible), acute neutrophilic hepatitis with icterus, vomiting, anorexia, diarrhea, fever, hemolytic anemia, urticaria, polyarthritis, facial swelling, polydipsia, polyuria and cholestasis.
2) Potentiated sulfonamides may cause hypothyroidism in the dog, particularly with extended therapy.
3) Acute hypersensitivity reactions manifesting as Type I (anaphylaxis) or Type III reaction (serum sickness) can also be seen. Hypersensitivity reactions appear to be more common in large breed dogs; Doberman Pinschers may possibly be more susceptible to this effect than other breeds.
4) Adverse effects noted in cats may include anorexia, leukopenias and anemias.
5) Sulfonamides (or their metabolites) can precipitate in the urine, particularly when given at high dosages for prolonged periods. Acidic urine or highly concentrated urine may also contribute to increased risk of crystalluria, hematuria and renal tubule obstruction.
6) Oral therapy has resulted in diarrhea development in some horses. Previous administration of potentiated sulfas has been implicated in increasing the mortality rate of equine diarrhea.

Environmental Property
None has been mentioned.

Symptoms and Treatment for Overdosage, and Antidote(s):
1) Manifestations of an acute overdosage can include symptoms of GI distress (nausea, vomiting, diarrhea), CNS toxicity (depression, headache, and confusion), facial swelling, bone marrow depression and increases in serum aminotransferases.
2) Oral ovedoses can be treated by emptying the stomach, (following usual protocols), and initiating symptomatic and supportive therapy. Acidification of the urine may be increase the renal elimination of trimethoprim, but could also cause sulfonamide crystalluria, particularly with sulfadiazine containing products. Complete blood count (and other laboratory parameters) should be monitored as necessary. Bone marrow suppression associated with chronic overdoses may be treated with folic acid (leucovorin) if severe. Peritoneal dialysis is not effective in removing TMP or sulfas from the circulation.

Shelf-Life:
3 years.
Used within 24 hours after reconstitution

Storage Condition(s):
Store at temperature below 30°C.

Product Description and Packing(s):
A beige to light brown suspension.
Plastic Bottle of 500ml, 1L

For food producing animals product:
Withdrawal period:
Chickens, and turkeys: Meat: 14days
Egg: Do not use in birds producing eggs for human consumption.

Maximum Residual Limit (MRL):
Sulfonamides: Cattle, sheep, goat: Milk 100µg/kg
Sulfadiazine: Poultry (muscle, fat, liver, kidney, edible offal) 0.1µg/kg
Poultry egg: 0.02µg/kg
Trimethoprim: All food producing species except horses: Muscle, fat, liver, kidney, milk 50µg/kg
Horses: Muscle, fat, liver, kidney 100µg/kg

Disposal: Dispose of any unused product or containers in accordance with guidance of the local waste regulation authority.



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